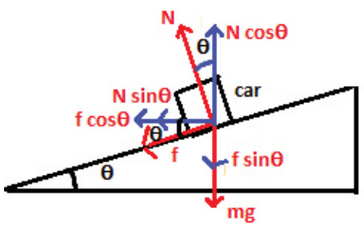
14. In the section 3.7.3 (Banking of road) we have not included the friction exerted by the road on the car. Suppose the coefficient of static friction between the car tyre and the surface of the road is μ_s, calculate the minimum speed with which the car can take safe turn?

When the car takes turn in the banked road, the following three forces act on the car.

(1) The gravitational force mg acting downwards
(2) The normal force N acting perpendicular to the surface of the road
(3) The static frictional force f acting on the car along the surface.

The following figure shows the forces acting on the horizontal and vertical direction.



When the car takes turn with the speed v, the centripetal force is exerted by horizontal component of normal force and static frictional force. It is given by

$$N\sin\theta + f\cos\theta = \frac{mv^2}{r} \qquad (1)$$

In the vertical direction, there is no acceleration. It implies that the vertical component of normal force is balanced by downward gravitational force and downward vertical component of frictional force. This can be expressed as

$$N\cos\theta = mg + f\sin\theta$$

Or
$$N\cos\theta - f\sin\theta = mg \qquad (2)$$

Diving the equation (1) by equation (2), we get

$$\frac{N\sin\theta + f\cos\theta}{N\cos\theta - f\sin\theta} = \frac{v^2}{rg} \qquad (3)$$

To calculate the maximum speed for the safe turn, we can use the maximum static friction is given by $f = \mu_s N$. By substituting this relation in equation (3), we get

$$\frac{N\sin\theta + \mu_s N\cos\theta}{N\cos\theta - \mu_s N\sin\theta} = \frac{v_{max}^{\ 2}}{rg}$$

By simplifying this equation,

$$\frac{N\cos\theta\left\{\left(\frac{N\sin\theta}{N\cos\theta}\right)+\mu_s\right\}}{N\cos\theta\left(1-\mu_s\frac{N\sin\theta}{N\cos\theta}\right)} = \frac{v_{max}^{\ 2}}{rg}$$

$$\frac{(\tan\theta+\mu_s)}{1-\mu_s\tan\theta} = \frac{v_{max}^{\ 2}}{rg}$$

The Maximum speed for safe turn is given

by $v_{max} = \sqrt{rg\frac{(\tan\theta+\mu_s)}{(1-\mu_s\tan\theta)}}$ (4)

Suppose we neglect the effect of friction ($\mu_s = 0$), then safe speed

$v_{safe} = \sqrt{rg\tan\theta}$ (5)

Note that the maximum speed with which the car takes safe turn is increased by friction (equation (4)). Suppose the car turns with speed $v < v_{safe}$, then the static friction acts up in the slope to prevent from inward skidding.

Appendix-1.indd 280 06-01-2022 17:55:22